

Confusing Pairs: Seemab ke liye MCQ Practice Sheet

Chemistry, Physics, Maths aur Biology · 9 September 2026

Seemab, ye sheet tumhare apne papers se banayi gayi hai.

Maine tumhare saare test dobara dekhe. Ek baat bohot clear nazar aayi: tumhare MCQs mein jo ghaltiyan hoti hain, wo "yaad nahi tha" wali ghaltiyan nahi hain. Zyada tar ghaltiyan **do aisi cheezon ke darmiyan hoti hain jo aapas mein bohot milti julti hain**, aur tum galti se doosri wali chun leti ho.

Misal ke taur par, Chemistry exam-20 mein tum ne **ek hi ghalti do baar** ki: oxidizing aur reducing agent ulta kar diya. Phir group rule aur period rule bhi do baar mila diya. Jab koi ghalti ek hi paper mein do baar ho, to iska matlab lapervahi nahi hai. Iska matlab hai ke dimagh mein wo cheez ulti baithi hui hai.

Isi liye in cheezon ko alag alag yaad karna kaam nahi karta. Jab tum sirf "electron affinity" yaad karti ho, to wo theek lagta hai. Lekin exam mein jab dono options saamne hote hain, tab confusion hoti hai. Inko **jori (pair) ki tarah, ek saath, ek doosre ke muqablay mein** yaad karna hoga.

Neeche har jori tumhare apne test ki ghalti se li gayi hai. Pehle inko parho, phir aakhir mein 40 MCQs hal karo.

CHEMISTRY

1. Electron Affinity vs Ionization Energy

Electron Affinity	Ionization Energy
Atom electron leta hai Energy release hoti hai (bahar aati hai)	Atom se electron nikalna parta hai Energy deni parti hai (kharch hoti hai)

Yaad rakhne ka tareeqa: "Affinity" ka matlab hai pasand karna. Atom electron ko pasand karta hai, isliye **leta** hai. Ionization mein **ion** banta hai, yani electron **jaata** hai.

2. Oxidizing Agent vs Reducing Agent

Oxidizing Agent	Reducing Agent
Electron leta hai Khud reduce hota hai	Electron deta hai Khud oxidize hota hai

Sab se aasan tareeqa: Agent hamesha **doosre ke saath wo karta hai jo apne saath nahi hota**. Oxidizing agent doosre ko oxidize karta hai, isliye khud reduce hota hai. Ulta socho, hamesha theek hoga.

3. Group Number vs Period Number (electronic configuration se)

Period Number	Group Number
Shells ki tayyad Sab se bara n (energy level)	Valence electrons se milta hai p-block ke liye: valence electrons + 10

Misal: $1s^2 2s^2 2p^2 \rightarrow$ sab se bara shell = 2, to **Period 2**. Valence electrons = 4, aur ye p-block hai, to $4 + 10 =$ **Group 14**. Sirf "Group 4" likhna ghalat hai.

4. Bohr Model vs Quantum Mechanical Model

Bohr Model (purana)	Quantum Mechanical Model (naya)
Electron fixed circular orbits mein ghoomta hai Jagah bilkul maloom hai	Electron orbitals mein hota hai Sirf probability maloom hai, exact jagah nahi

Farq: Orbit = raasta (line). Orbital = ilaqa (area) jahan electron milne ka imkaan hai.

5. Giant Covalent vs Simple Molecular

Giant Covalent (network)	Simple Molecular
Poora crystal ek hi barra network Melting point bohot zyada Misal: diamond, SiO_2	Chhote chhote alag molecules Weak forces beech mein Melting point kam Misal: iodine, ice

Sawal ka trick: Agar melting point bohot zyada hai to **giant covalent**. Diamond isliye nahi pighalta kyunke poora crystal ek hi bara molecule hai.

6. Catalyst kya karta hai aur kya NAHI karta

Catalyst ye KARTA hai	Catalyst ye NAHI karta
Naya raasta deta hai jis mein activation energy kam ho	ΔH change NAHI karta

Reaction tez kar deta hai	Khud khatam NAHI hota Temperature NAHI barhata
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Ye bohot zaroori hai. Catalyst sirf pahar ke ooper se jaane ke bajaye **surang (tunnel)** bana deta hai. Shuru aur aakhir ki jagah wahi rehti hai, isliye ΔH bilkul wahi rehta hai.

7. Metallic Bond vs Ionic Bond

Metallic Bond	Ionic Bond
Metal atoms + free (delocalized) electrons ka samundar Bijli conduct karta hai solid mein bhi	Metal ne electron diya, non-metal ne liya Bijli sirf molten ya solution mein

8. Empirical Formula vs Molecular Formula

Empirical Formula	Molecular Formula
Sab se chhota ratio Misal: CH_2O	Asli tayadaad Empirical $\times n$ Misal: $\text{C}_6\text{H}_{12}\text{O}_6$

Jab sawal mein mass ya moles poochha jaye, to **molecular formula** use karo, empirical nahi. Ye ghalti exam-18 mein hui thi.

PHYSICS

9. Speed-Time Graph: Gradient vs Area

Gradient (dhalan)	Area (neech ka hissa)
= Acceleration Unit: m/s^2	= Distance (kitna safar tay hua) Unit: m

Ye tumhari sab se purani ghalti hai, kai papers mein aa chuki hai. Yaad karne ka tareeqa: area ek **jagah** ghairti hai, aur distance bhi **jagah** hai. Gradient tezi dikhata hai, aur acceleration bhi tezi hai.

10. Distance vs Displacement, Speed vs Velocity

Distance / Speed	Displacement / Velocity
Scalar: sirf kitna Poora raasta ginta hai	Vector: kitna + kis taraf Sirf seedhi doori shuru se aakhir tak

11. Mass vs Weight

Mass	Weight
Maadde ki miqdaar, kg Har jagah same Physical balance se	$W = mg$, Newton Jagah badalne se badalta hai Spring balance se

Chaand par mass wahi rahegi, weight kam ho jayega kyunke g kam hai (1.62 m/s^2).

MATHS

12. $\sqrt{a} + \sqrt{b}$ ka matlab $\sqrt{a+b}$ NAHI hota

GHALAT	SAHIH
$\sqrt{48} + \sqrt{27} = \sqrt{75}$	$\sqrt{48} = 4\sqrt{3}$ aur $\sqrt{27} = 3\sqrt{3}$ To $4\sqrt{3} + 3\sqrt{3} = 7\sqrt{3}$

Pehle har radical ko simple karo, **phir** jama karo. Radical ke andar wale numbers seedha jama nahi hotay.

13. Absolute Value: $|-8| = 8$

Do khari lakeerein **bracket nahi hain**. Inka matlab hai **zero se faasla**, aur faasla kabhi minus nahi hota. Tumhari kitaab mein Unit 5 §5.3.1 mein likha hai: " $|5| = 5$ aur $|-5| = 5$ ". To $|-8| + |3| = 8 + 3 = 11$, minus 5 nahi.

14. Square root ke saath $\pm$ lagana

Division method se square root nikalne ke baad hamesha $\pm$ likho. Tumhari kitaab Unit 4 Example 29 mein likha hai: " $\therefore$ Square root $= \pm(3x - 7y)$ ". Sirf $(3x^2 - 2x + 1)$ likhna adhoora hai.

15. Inequality mein DONO taraf kaam karo

GHALAT	SAHIH
$-3 \leq 2x - 1 \leq 5 \rightarrow [1, 3]$	Teenon hisson mein 1 jama karo: $-2 \leq 2x \leq 6$ Teenon ko 2 se taqseem: $-1 \leq x \leq 3$

Jo kaam beech mein karo wahi **dono kinaron** par bhi karo. Minus ki alamat kabhi mat girao.

16. Power/Index ki ghaltiyan

GHALAT	SAHIH
$(2x)^2 = 2x^2$ $(5y)^2 = 5y^2$ $2^{-3} = -8$	$(2x)^2 = 4x^2$ $(5y)^2 = 25y^2$ $2^{-3} = 1/8$

Bracket ke andar jo bhi hai, **poora** square hota hai, sirf x ya y nahi. Aur minus power ka matlab **ulta karo**, minus nahi banao.

BIOLOGY

17. Purines vs Pyrimidines

Purines	Pyrimidines
Do ring (bara) Adenine, Guanine	Ek ring (chhota) Cytosine, Thymine, Uracil

Yaad karne ka tareeqa: **Purines** mein **Pure Aur G** (A aur G). Naam lamba, ring bhi zyada.

18. Mitosis vs Meiosis

Mitosis	Meiosis
2 cells bantay hain Chromosomes same (diploid) Body/somatic cells ke liye	4 cells bantay hain Chromosomes aadhay (haploid) Gametes ke liye, variation deta hai

19. Somatic Cell vs Gamete

Somatic ka matlab **jism ka koi bhi cell** hai, sirf "animal cell" nahi. Ye lafz exam-09 mein ghalat likha gaya tha.

20. Equatorial Plate

Spelling: **equatorial** (equator se, yani darmiyani lakeer), "equilatoral" nahi. Metaphase mein chromosomes cell ke beech mein isi plate par lagtay hain.

AB PRACTICE KARO: 40 MCQs

Har sawal upar wali kisi jori se hai. Har sawal mein **dhoka dene wala option** jaan boojh kar rakha gaya hai, bilkul asli exam ki tarah. Rough kaam zaroor likho, sirf dimagh mein hal mat karo.

1. Electron affinity ka matlab hai:

(a) Electron lene par release hone wali energy (b) Electron nikalne ke liye zaroori energy (c) Bond torne ki energy (d) Pighalne ki energy

2. Ionization energy hamesha:

(a) Release hoti hai (b) Zero hoti hai (c) Deni parti hai (d) Negative hoti hai

3. Reaction mein oxidizing agent wo hai jo:

(a) Electron deta hai aur khud oxidize hota hai (b) Tabdeel nahi hota (c) Sirf non-metal hota hai (d) Electron leta hai aur khud reduce hota hai

4. $Mg + Cl_2 \rightarrow MgCl_2$ mein reducing agent kaun hai?

(a) Cl_2 (b) $MgCl_2$ (c) Mg (d) Koi nahi

5. Ek element ki configuration $1s^2 2s^2 2p^4$ hai. Iska period hai:

(a) 2 (b) 4 (c) 6 (d) 16

6. Isi element ($1s^2 2s^2 2p^4$) ka group hai:

(a) 4 (b) 6 (c) 2 (d) 16

7. Quantum mechanical model ke mutabiq electron ki jagah bayan hoti hai:

(a) Probability wale orbitals se (b) Fixed circular orbits se (c) Nucleus par thehre hue point se (d) Seedhi lakeer se

8. "Orbit" aur "orbital" mein farq ye hai ke orbital:

(a) Ek mutayyan raasta hai (b) Ek ilaqa hai jahan electron milne ka imkaan hai (c) Sirf hydrogen mein hota hai (d) Nucleus ka hissa hai

9. Diamond ka melting point bohot zyada hai kyunke:

(a) Chhote molecules weak forces se jure hain (b) Ye ionic lattice hai (c) Ye giant covalent network hai (d) Is mein free electrons hain

10. In mein se kaun sa simple molecular structure hai?

(a) Diamond (b) Silicon dioxide (c) Iodine (d) Sodium chloride

11. Catalyst reaction tez karta hai kyunke wo:

(a) Temperature barha deta hai (b) ΔH barha deta hai (c) Kam activation energy wala raasta deta hai (d) Reaction mein khatam ho jata hai

- 12.** Catalyst dalne se reaction ka ΔH :
(a) Barh jata hai (b) Kam ho jata hai (c) Zero ho jata hai (d) Bilkul wahi rehta hai
- 13.** Metal bijli solid halat mein bhi conduct karta hai kyunke us mein:
(a) Free ghoomne wale electrons hain (b) Ions hain jo hil saktay hain (c) Covalent bonds hain (d) Weak forces hain
- 14.** Ionic compound bijli kab conduct karta hai?
(a) Hamesha (b) Sirf solid mein (c) Kabhi nahi (d) Sirf molten ya solution mein
- 15.** Ek compound ka empirical formula CH_2O hai aur $M_r = 180$. Molecular formula hai:
(a) CH_2O (b) $\text{C}_2\text{H}_4\text{O}_2$ (c) $\text{C}_6\text{H}_{12}\text{O}_6$ (d) $\text{C}_3\text{H}_6\text{O}_3$
- 16.** Moles nikalne ke liye kaun sa formula use hoga?
(a) Empirical formula ka mass (b) Molecular formula ka mass (c) Dono barabar (d) Sirf atomic number
- 17.** Speed-time graph ka **gradient** kya deta hai?
(a) Distance (b) Time (c) Force (d) Acceleration
- 18.** Speed-time graph ke neeche ka **area** kya deta hai?
(a) Acceleration (b) Speed (c) Distance (d) Momentum
- 19.** Ek speed-time graph seedhi ufqi (horizontal) lakeer hai. Iska matlab:
(a) Acceleration zero hai, speed constant hai (b) Body ruki hui hai (c) Speed barh rahi hai (d) Distance zero hai
- 20.** Distance aur displacement mein farq ye hai ke displacement:
(a) Hamesha bara hota hai (b) Scalar hai (c) Vector hai, direction bhi rakhta hai (d) Kabhi zero nahi hota
- 21.** Chaand par le jaane se kya badlega?
(a) Mass (b) Weight (c) Dono (d) Koi nahi
- 22.** Zameen par 600 N wazan wale shakhs ka chaand par wazan ($g = 1.62 \text{ m/s}^2$) hoga taqreeban:
(a) 600 N (b) 972 N (c) 60 N (d) 99 N
- 23.** Mass naapne ka aala hai:
(a) Spring balance (b) Physical balance (c) Stopwatch (d) Measuring cylinder
- 24.** $\sqrt{50} + \sqrt{18}$ barabar hai:
(a) $8\sqrt{2}$ (b) $\sqrt{68}$ (c) 68 (d) $2\sqrt{17}$
- 25.** $\sqrt{12} + \sqrt{27}$ barabar hai:
(a) $\sqrt{39}$ (b) $5\sqrt{3}$ (c) 39 (d) $6\sqrt{3}$
- 26.** $|-9| + |4| =$
(a) -13 (b) 13 (c) -5 (d) 5
- 27.** $|-6| - |-2| =$
(a) -8 (b) -4 (c) 8 (d) 4
- 28.** $9x^2 - 12x + 4$ ka square root hai:
(a) $3x - 2$ (b) $3x + 2$ (c) $\pm(3x + 2)$ (d) $\pm(3x - 2)$
- 29.** Agar $-5 \leq 2x - 1 \leq 7$ ho to x ka set hai:
(a) $[2, 4]$ (b) $[-2, 4]$ (c) $[-2, 8]$ (d) $[-3, 3]$
- 30.** Agar $-7 < 3x + 2 \leq 11$ ho to x ka set hai:
(a) $(-3, 3]$ (b) $(3, 3]$ (c) $(-9, 9]$ (d) $(-5, 13]$
- 31.** $(3x)^2$ barabar hai:
(a) $3x^2$ (b) $6x^2$ (c) $9x^2$ (d) $9x$
- 32.** $(4y)^2$ barabar hai:
(a) $4y^2$ (b) $8y^2$ (c) $16y^2$ (d) $16y$
- 33.** 3^{-2} barabar hai:
(a) -9 (b) -6 (c) 9 (d) $1/9$
- 34.** 5^{-1} barabar hai:
(a) -5 (b) $1/5$ (c) 5 (d) $-1/5$
- 35.** Purines mein shamil hain:
(a) Adenine aur Guanine (b) Cytosine aur Thymine (c) Thymine aur Uracil (d) Cytosine aur Guanine
- 36.** Pyrimidine ki pehchan hai:
(a) Do ring (b) Ek ring (c) Teen ring (d) Koi ring nahi
- 37.** Mitosis ke baad kitne cells bantay hain?
(a) 2, chromosomes same (b) 4, chromosomes aadhay (c) 2, chromosomes aadhay (d) 4, chromosomes same
- 38.** Meiosis ki khaas baat hai:
(a) Do diploid cells (b) Chromosomes double (c) Sirf body cells mein (d) Chahr haploid cells aur variation
- 39.** Somatic cell ka matlab hai:
(a) Sirf animal cell (b) Jism ka koi bhi cell (gamete ke ilawa) (c) Sirf plant cell (d) Sirf gamete
- 40.** Metaphase mein chromosomes lagtay hain:
(a) Equatorial plate par (b) Poles par (c) Nucleus ke andar (d) Cell wall par

luck!